

Composite Design Pattern

In this activity, I learned how the Composite Design Pattern is used to manage hierarchical structures, such as forum threads containing posts and replies. The original implementation used `instanceof` conditions to identify whether an object was a post or a thread. This made the rendering process dependent on specific classes and could cause problems when another component, such as a poll, was added to the forum structure.

To solve this problem, I created a common `ForumComponent` interface that is implemented by both the `Post` and `Thread` classes. I completed the `Thread::display()` method by looping through its child components and calling their `display()` methods recursively. I also created the RAG variant using `Document`, `Section`, and `Chunk`, with methods such as `getText()` and `embed()`. In addition, I added a new `Bundle` component to demonstrate that another component could be introduced without changing the client code.

Through this activity, I understood how the Composite Pattern allows individual objects and groups of objects to be treated uniformly. The recursive structure makes it easier to display and manage nested content because each component handles its own behavior. I also learned that the same pattern can be applied to different systems, such as forums and document-processing applications. This makes the design reusable and suitable for applications with part-whole relationships.